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MATHEMATICS
Paper 1
July./ Aug.2026
$2\frac{1}{4}$ hours

BOARD OF GOVERNORS EXAMINATION CONSULTS

Uganda Certificate of Education

MATHEMATICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES

*This examination paper has **two** Sections; **A** and **B**. it consists of **six** items*

*Section **A** has **two compulsory** items*

*Section **B** has **two** parts **I** and **II**. Respond to **one** item from each part*

*Respond to **four** examination items in all*

*Any additional item(s) respond to will **not** be scored*

***All** responses **must** be written in the answer booklet(s) provided.*

Graph Paper is provided

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Turn Over

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SECTION A

*Attempt **both** items in this section*

ITEM 1

Joseph has decided to start up his business of phone repair and sell of accessories. His friend has advised him that he will need four million two hundred fifteen thousand Ugandan shillings to start. He has to pay 3 months' rent for a room that is charged at UGX 250,000 per month as a new tenant. After paying rent, he is to use $^1/_5$ of the remaining amount to pay for equipment repairs, and reserve 15% of the remainder for the operation license; then use the rest for stocking phone accessories. Joseph needs your help to allocate the money as planned.

A friend has promised to give him a loan at a 10% interest at the time of paying back the money.

Meanwhile he got a client who had forgotten the 4 – digit PIN – code for the mobile money line, however the client remembered that he had saved it starting with a zero, and the other 3 digits can be got by converting 51 to septenary base in order to get the actual 4 – digit PIN – code, so Joseph was asked to help get the actual pin.

Tasks

Help Joseph to determine;

(a) the actual allocation of the money as planned,

(b) how much money he is supposed to return when paying back his friend,

(c) the actual 4 – digit PIN – code for the client's mobile money line.

ITEM 2

The O' level Entrepreneurship students in a certain school want to make buns and cakes for sale during their school visitation day. They plan to sell each bun at Shs.500 and each cake at Shs.1000. To maximize sales, they plan to use at most 4000g of baking flour. Making each bun consumes 80g and a cake consumes 50g of the baking flour. They do not want the number of cakes to be more than twice the buns and the total number of cakes and buns produced should be at least 50.

According to their projections, they expect to earn a percentage profit of 40% in their first production having incurred a total cost of Shs.250,000 for their first production.

Tasks

(a) With aid of relevant expressions and inequalities, show the feasible region and recommend the number of buns and cakes the students can make to achieve their goal.

(b) According to their projections, what will be their profit in the first production?

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SECTION B

PART I

*Respond to **one** item only in this part*

ITEM 3

In a bid to improve water security in a local village, a committee surveyed 130 households regarding their reliance on clean water sources, which included **T**ap water, **B**orehole water, **R**ainwater, and other alternative sources. From the survey findings, 83 households used Tap water (with 18 using Tap water only), 62 households used Borehole water, and 60 households used Rainwater. Furthermore, 30 households used both Tap and Borehole water, 40 used both Tap and Rainwater, while 20 used both Borehole and Rainwater.

The committee established that a water shortage will occur in the village if more than half of the total households rely on just one of the three clean water sources.

To promote clean multi – source water harvesting, an organization decided to reward households that utilize all the three clean water sources through a game where a representative draws two marbles at random without replacement from a box containing 4 red marbles and 7 green marbles and wins a prize if the drawn marbles are of different colors.

Tasks

(a) Determine if the village will experience a water shortage.

(b) Calculate the probability that a qualifying household wins the game.

ITEM 4

A local hospital recorded the body weights in kilograms of patients admitted during a certain week as follows:

51	**56**	**67**	**30**	**21**	**42**	**34**	**38**	**83**	**38**	**75**	**45**	**34**	**38**	**58**	**50**
56	**76**	**91**	**87**	**25**	**56**	**27**	**64**	**61**	**92**	**90**	**64**	**40**	**49**	**77**	**54**
58	**46**	**75**	**52**	**60**	**66**	**22**	**94**	**44**	**52**	**34**	**64**	**72**	**80**	**52**	**44**

The medical superintendent needs to determine the average patient weight and estimate the weight common to most patients in order to guide medical dosage planning and facility resource allocation.

Tasks

(a) Summarize the data in groups of 10 starting with a weight of 20 in a suitable table, and use your summary to calculate the average weight of the patients.

(b) Help the superintendent to estimate the weight shared by most patients.

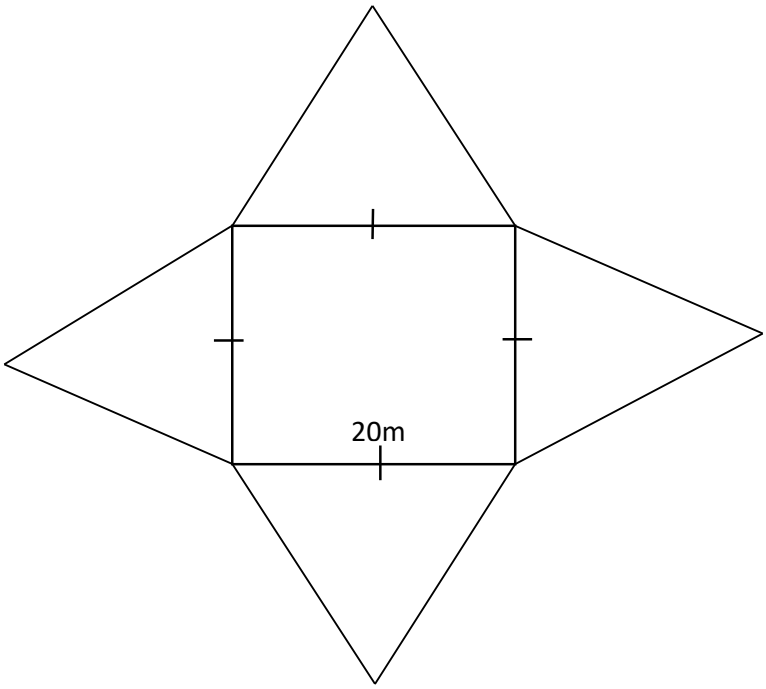
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PART II

*Respond to **one** item only in this part*

ITEM 5

Your brother has secured a new job as a junior engineer at a construction firm that specializes in temporary shelter designs. His manager provided him with a flat 2 Dimensional net model of a pyramid – shaped canvas tent for a disaster-relief project and asked him to assemble a 3 Dimensional structural display for clients, but he requires your technical assistance.



He needs to determine the minimum canvas material required to construct one tent whose square base measures 20 metres on each side and stands at a vertical height of 24 metres from the center of the base.

On his payroll terms, he earns a gross monthly salary of UGX.1,500,000, which includes a non-taxable transport allowance of UGX.300,000 and a lunch allowance of UGX.7,000 per day for the 5 days he works each week (assuming 4 weeks a month). He plans to open an education insurance policy for his child requiring UGX.250,000 monthly and save UGX.400,000 for future investments, subject to the Pay As You Earn (PAYE) tax rates shown below:

Taxable Income (UGX)	Tax Rate (%)
1 – 235,000	0
235,001 – 335,000	10
335,001 – 410,000	20
410,001 – 1,500,000	30

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Tasks

(a) (i) By sketching show how the tent will look after it is assembled from the net.

(ii) Using mathematical calculations, determine the minimum canvas material required to construct the tent structure.

(b) Do you advise him to open the child insurance policy? Justify your response with financial calculations.

ITEM 6

A tourist on a safari in a national park set out from the main lodge at Point A to explore the park's wildlife sites. She first traveled along a track on a bearing of S30°E for a distance of 5 km to reach a crater lake view site at Point B. From Point B, she changed direction and traveled on a bearing of 235° for 9 km to a river campsite at Point C. To avoid driving back through the same long route before sunset, she decided to travel directly from Point C back to the main lodge at Point A, but needed to determine the exact direction and distance for a direct return. For the entire journey, she hired a safari vehicle whose owner charged UGX.500,000, which was meant to cover vehicle hire and fuel for the total distance to be covered. The vehicle consumes 2 litres per km, fuel costs UGX.6,550 per litre at the fuel stations, and the driver receives a 5% discount for every UGX.100,000 worth of fuel purchased as a regular customer.

Tasks

(a) Determine the direction and distance from Point C to the main lodge at Point A that should be shown on the tourist route map.

(b) (i) Determine the actual total cost spent on fuel for the entire journey.

(ii) Determine whether the vehicle owner will save some money from the UGX.500,000 provided, and if so, calculate how much.

END

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